

Answer **all** questions.

1. Biological control is used to reduce the numbers of a pest population through the introduction of another species. It has been used with varying success since the 19th century.

(a) (i) State **two** advantages of this method of control. [2]

Advantage 1

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Advantage 2

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(ii) State **two** disadvantages of this method of control. [2]

Disadvantage 1

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Disadvantage 2

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(b) The whitefly (*Trialeurodes vaporariorum*) is a pest which damages greenhouse crops such as tomatoes. Whitefly numbers can be reduced by using the biological control agent *Encarsia formosa*.

***Encarsia formosa* fact file**

- *E. formosa* is a tiny wasp that lays eggs inside developing whitefly.
- When the eggs hatch, the young wasps kill the developing whitefly from the inside.
- Optimal conditions for *E. formosa* are temperatures over 20°C.
- When daytime temperatures are less than 17°C, *E. formosa* activity is significantly reduced, making it less effective.

